

SANATAN SAATHI

Technical Development Blueprint

Complete Architecture, Database Schema, API Design & Development Plan

Version 1.0

January 2026

Prepared by E1 Agent | Emergent Labs

Based on SanatanSaathi Blueprint v3.0

Sanatan Saathi — Technical Development Blueprint

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1. PROJECT UNDERSTANDING & VISION

What is Sanatan Saathi?

Sanatan Saathi is a **comprehensive digital spiritual companion for Sanatan Dharma** — a platform where devotees can:

- **Read & Listen** to Vedic Mantras, Chalisas, Ashtakams, Artis, Kathas in 12+ Indian languages
- **Learn** through Beginner/Expert modes with word-by-word breakdown, transliteration, and meanings
- **Practice** daily spiritual routines with personalized mantra lists
- **Ask** scripture-based questions to an AI agent (VedaChat powered by Claude)
- **Create & Share** beautiful shloka video cards on social media
- **Explore** sacred texts (Bhagavad Gita, Ramayana, Mahabharata, Vedas, Puranas)

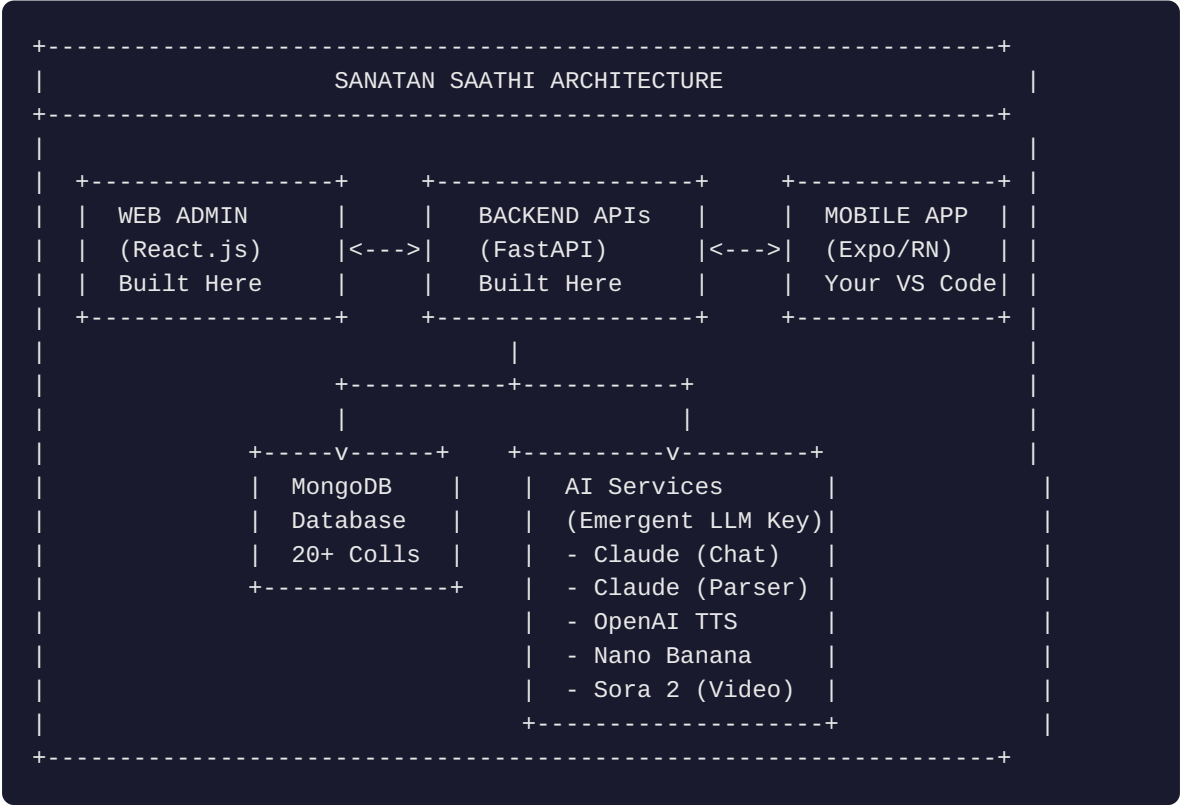
Core Value Proposition

- One app for ALL spiritual content (not scattered across multiple apps)
- AI-powered learning (VedaChat answers from actual scriptures with verse references)
- Beginner-friendly with Expert depth
- 12+ language support with Sanskrit original + transliteration + meaning
- Daily routine builder for consistent spiritual practice

Target Users

User Type	Description	Key Features
Beginner Devotee	New to scriptures, wants to learn basics	Beginner mode, word breakdown, slow TTS, daily routine
Expert Practitioner	Regular practitioner, knows shlokas	Expert mode, continuous play, minimal annotations
Content Admin	Manages spiritual content	Admin panel, DOCX upload, verse editing
Scholar/ Researcher	Deep study of Vedas/Puranas	VedaChat AI, Vedas & Puranas section

2. ARCHITECTURE OVERVIEW



Tech Stack Decisions

Layer	Technology	Why
Web Admin Frontend	React.js + Tailwind CSS + Radix UI	Modern, component-based, responsive admin panel
Backend API	FastAPI (Python)	Fast, async, auto-docs (Swagger), great for AI integrations
Database	MongoDB	Document-based, flexible schema for varied content types (mantras, kathas, artis have different structures). Designed for easy migration to Supabase/PostgreSQL later
AI Chat	Claude (via Emergent Key)	Best for nuanced scripture understanding, multi-turn conversations
AI Parsing	Claude (via Emergent Key)	Excellent at extracting structured data from DOCX/PDF
Text-to-Speech	OpenAI TTS (via Emergent Key)	High-quality voices for Hindi/Sanskrit, no separate API key needed
		Shloka card backgrounds, decorative images

Layer	Technology	Why
Image Generation	Gemini Nano Banana (via Emergent Key)	
Video Generation	Sora 2 (via Emergent Key)	Short shloka video reels
Authentication	JWT (JSON Web Tokens)	Stateless, works across web + mobile

3. WHAT WE BUILD HERE vs EXPO MOBILE

Built on This Platform (Web Admin + Backend)

Everything the mobile app needs to function:

Component	Status	Description
Complete REST API	Built Here	All endpoints for content, auth, user, admin
Admin Panel	Built Here	Full web dashboard for content management
Database Schema	Built Here	All collections with sample data
VedaChat AI	Built Here	Claude-powered chatbot API
DOCX Parser	Built Here	Upload & parse pipeline
PDF Parser	Built Here	Vedas/Puranas extraction
TTS Generation	Built Here	Audio generation API
Image Generation	Built Here	Shloka card API
Auth System	Built Here	JWT-based, works with mobile

Built in Your VS Code (Expo Mobile App)

The mobile app consumes our APIs:

Component	Where	What You Need
Expo Project Setup	Your VS Code	Expo CLI, React Native
5-Tab Navigation	Your VS Code	expo-router
Home Screen	Your VS Code	Calls our <code>/api/home/*</code> endpoints
Vedic Mantras Screens	Your VS Code	Calls our <code>/api/content/*</code> endpoints
Divya Granth Screens	Your VS Code	Calls our <code>/api/granth/*</code> endpoints
Vedas & Puranas Screens	Your VS Code	Calls our <code>/api/vedas/*</code> endpoints
VedaChat Screen	Your VS Code	Calls our <code>/api/vedachat/*</code> endpoints

Component	Where	What You Need
Audio Player	Your VS Code	expo-av + our audio URLs
Profile/Settings	Your VS Code	Calls our <code>/api/user/*</code> endpoints

I will provide **complete Expo integration guide** with code snippets for each screen.

4. DATABASE SCHEMA (MongoDB Collections)

4.1 Content Collections

`content_items` — Master Content Table

```
{
  "_id": "ObjectId",
  "category": "chalisa | vedic_mantra | ashtakam | sahasranama |
nama_ramayanam",
  "slug": "hanuman-chalisa",
  "title_hi": "हनुमान चालीसा",
  "title_en": "Hanuman Chalisa",
  "title_sa": "हनुमान् चालीसा",
  "deity": "Hanuman",
  "deity_hi": "हनुमान",
  "description_hi": "...",
  "description_en": "...",
  "thumbnail_url": "https://...",
  "audio_url": "https://...",
  "audio_duration_seconds": 420,
  "has_beginner_mode": true,
  "has_expert_mode": true,
  "total_verses": 40,
  "sort_order": 1,
  "is_active": true,
  "is_premium": false,
  "tags": ["daily", "popular", "hanuman"],
  "supported_languages": ["hi", "en", "sa", "mr", "gu", "ta", "te", "bn",
"kn", "ml", "pa", "od"],
  "like_count": 0,
  "created_at": "2026-01-01T00:00:00Z",
  "updated_at": "2026-01-01T00:00:00Z",
  "created_by": "admin_user_id",
  "status": "draft | published | archived"
}
```

`content_verses` — Verses/Shlokas

```
{
  "_id": "ObjectId",
  "item_id": "content_item_id (ref)",
  "verse_num": 1,
```

```

"verse_type": "doha | chaupai | shloka | mantra | stanza",
"sanskrit_text": "श्रीगुरु चरन सरोज रज...",
"transliteration": "Shri Guru Charan Saroj Raj...",
"audio_url": "https://... (individual verse audio)",
"audio_start_ms": 0,
"audio_end_ms": 15000,
"sort_order": 1,
"is_active": true
}

```

verse_meanings — Multilingual Meanings

```

{
  "_id": "ObjectId",
  "verse_id": "content_verse_id (ref)",
  "language": "hi | en | mr | gu | ta | te | bn | kn | ml | pa | od",
  "meaning": "गुरु के चरण कमल की धूल से...",
  "word_breakdown": [
    {"word": "श्रीगुरु", "meaning_hi": "श्री गुरु", "meaning_en": "Revered Guru"},
    {"word": "चरन", "meaning_hi": "चरण", "meaning_en": "Feet"},
    {"word": "सरोज", "meaning_hi": "कमल", "meaning_en": "Lotus"},
    {"word": "रज", "meaning_hi": "धूल", "meaning_en": "Dust"}
  ]
}

```

4.2 Katha & Puja Collections

katha_items

```

{
  "_id": "ObjectId",
  "title_hi": "सत्यनारायण कथा",
  "title_en": "Satyanarayan Katha",
  "deity": "Vishnu",
  "deity_hi": "विष्णु",
  "thumbnail_url": "https://...",
  "intro_text_hi": "...",
  "intro_text_en": "...",
  "puja_vidhi": [
    {"step": 1, "text_hi": "स्नान करें", "text_en": "Take bath"},
    {"step": 2, "text_hi": "...", "text_en": "..."}
  ],
  "samagri": [
    {"item_hi": "अक्षत (चावल)", "item_en": "Rice", "quantity": "100g"},
    {"item_hi": "हल्दी", "item_en": "Turmeric", "quantity": "1 packet"}
  ],
  "total_chapters": 5,
  "sort_order": 1,
  "is_active": true,
}

```

```
"status": "published"
}
```

katha_verses

```
{
  "_id": "ObjectId",
  "katha_id": "katha_item_id (ref)",
  "chapter_num": 1,
  "chapter_name_hi": "प्रथम अध्याय",
  "chapter_name_en": "Chapter 1",
  "verse_num": 1,
  "text_hi": "...",
  "text_sa": "...",
  "meaning_hi": "...",
  "meaning_en": "...",
  "audio_url": "https://..."
}
```

4.3 Arti Collections

arti_items

```
{
  "_id": "ObjectId",
  "title_hi": "ॐ जय जगदीश हरे",
  "title_en": "Om Jai Jagdish Hare",
  "deity": "Vishnu",
  "deity_hi": "विष्णु",
  "thumbnail_url": "https://...",
  "music_url": "https://...",
  "audio_duration_seconds": 300,
  "sort_order": 1,
  "is_active": true,
  "status": "published"
}
```

arti_lines

```
{
  "_id": "ObjectId",
  "arti_id": "arti_item_id (ref)",
  "line_num": 1,
  "text_hi": "ॐ जय जगदीश हरे",
  "text_sa": "...",
  "transliteration": "Om Jai Jagdish Hare",
  "is_refrain": true,
  "audio_start_ms": 0,
}
```

```
"audio_end_ms": 8000
}
```

4.4 Divya Granth (Sacred Texts)

granth_books

```
{
  "_id": "ObjectId",
  "title_hi": "श्रीमद्भगवद्गीता",
  "title_en": "Bhagavad Gita",
  "slug": "bhagavad-gita",
  "description_hi": "...",
  "description_en": "...",
  "thumbnail_url": "https://...",
  "total_chapters": 18,
  "total_verses": 700,
  "sort_order": 1,
  "is_active": true
}
```

granth_chapters

```
{
  "_id": "ObjectId",
  "book_id": "granth_book_id (ref)",
  "chapter_num": 1,
  "title_hi": "अर्जुन विषाद योग",
  "title_en": "Arjuna Vishada Yoga",
  "description_hi": "...",
  "total_verses": 47,
  "shloka_audio_url": "https://... (male Sanskrit voice)",
  "meaning_audio_url": "https://... (female Hindi voice)",
  "sort_order": 1
}
```

granth_verses

```
{
  "_id": "ObjectId",
  "chapter_id": "granth_chapter_id (ref)",
  "verse_num": 1,
  "sanskrit": "धृतराष्ट्र उवाच...",
  "transliteration": "Dhritarashtra Uvacha...",
  "meaning_hi": "...",
  "meaning_en": "...",
  "meaning_mr": "...",
  "word_meanings": [
```

```

    {"word": "धृतराष्ट्र", "meaning": "Dhritarashtra"},
    {"word": "उवाच", "meaning": "said"}
  ],
  "audio_start_ms": 0,
  "audio_end_ms": 12000,
  "sort_order": 1
}

```

4.5 Vedas & Puranas

veda_books

```

{
  "_id": "ObjectId",
  "title_hi": "ऋग्वेद",
  "title_en": "Rig Veda",
  "category": "veda | purana",
  "sub_type": "rig | sama | yajur | atharva",
  "pdf_url": "https://...",
  "thumbnail_url": "https://...",
  "total_chapters": 10,
  "description_hi": "...",
  "description_en": "...",
  "sort_order": 1,
  "is_active": true,
  "parsing_status": "pending | processing | completed | error"
}

```

veda_chapters

```

{
  "_id": "ObjectId",
  "book_id": "veda_book_id (ref)",
  "chapter_num": 1,
  "title_hi": "प्रथम मण्डल",
  "title_en": "First Mandala",
  "total_verses": 191,
  "sort_order": 1
}

```

veda_verses

```

{
  "_id": "ObjectId",
  "chapter_id": "veda_chapter_id (ref)",
  "verse_num": 1,
  "text_sa": "अग्निमीळे पुरोहितं...",
  "transliteration": "Agnim Ile Purohitam...",
}

```

```

    "meaning_hi": "...",
    "meaning_en": "...",
    "audio_url": "https://...",
    "sort_order": 1
  }

```

4.6 User Collections

user_profiles

```

{
  "_id": "ObjectId",
  "name": "Rahul Sharma",
  "email": "rahul@example.com",
  "phone": "+919876543210",
  "avatar_url": "https://...",
  "language_pref": "hi",
  "mode_pref": "beginner | expert",
  "is_premium": false,
  "subscription_plan": "free | monthly | yearly",
  "subscription_expires_at": null,
  "streak_count": 15,
  "streak_last_date": "2026-01-15",
  "notification_enabled": true,
  "dark_mode": false,
  "tts_voice_pref": "male_hindi",
  "created_at": "2026-01-01T00:00:00Z",
  "last_active_at": "2026-01-15T08:30:00Z"
}

```

user_saved_shlokas

```

{
  "_id": "ObjectId",
  "user_id": "user_profile_id (ref)",
  "item_id": "content_item_id (ref)",
  "verse_id": "content_verse_id (ref)",
  "category": "chalisa | vedic_mantra | ...",
  "sort_order": 1,
  "added_at": "2026-01-15T08:30:00Z"
}

```

user_likes

```

{
  "_id": "ObjectId",
  "user_id": "user_profile_id (ref)",
  "item_id": "content_item_id (ref)",

```

```
{
  "liked_at": "2026-01-15T08:30:00Z"
}
```

user_reading_history

```
{
  "_id": "ObjectId",
  "user_id": "user_profile_id (ref)",
  "item_id": "content_item_id (ref)",
  "item_type": "content | granth | veda | katha",
  "last_verse_num": 15,
  "progress_percent": 37.5,
  "total_time_seconds": 600,
  "last_read_at": "2026-01-15T08:30:00Z"
}
```

user_videos

```
{
  "_id": "ObjectId",
  "user_id": "user_profile_id (ref)",
  "verse_id": "content_verse_id (ref)",
  "video_url": "https://...",
  "thumbnail_url": "https://...",
  "is_public": true,
  "created_at": "2026-01-15T08:30:00Z"
}
```

4.7 Admin & System Collections

admin_users

```
{
  "_id": "ObjectId",
  "email": "admin@sanatansaathi.com",
  "password_hash": "bcrypt_hash",
  "name": "Super Admin",
  "role": "super_admin | content_admin | moderator",
  "is_active": true,
  "created_by": "admin_user_id or null",
  "last_login_at": "2026-01-15T08:30:00Z",
  "created_at": "2026-01-01T00:00:00Z"
}
```

panchang

```
{
  "_id": "ObjectId",
  "date": "2026-01-15",
  "tithi": "शुक्ल पक्ष तृतीया",
  "nakshatra": "रोहिणी",
  "yoga": "शोभन",
  "karana": "बव",
  "sunrise": "07:12",
  "sunset": "17:48",
  "rahu_kaal": "10:30-12:00",
  "festival_name": "मकर संक्रांति",
  "festival_name_en": "Makar Sankranti",
  "is_panchak": false,
  "is_bhadra": false,
  "linked_content_ids": ["content_item_id_1", "content_item_id_2"],
  "special_message_hi": "...",
  "special_message_en": "..."
}
```

upload_logs

```
{
  "_id": "ObjectId",
  "admin_id": "admin_user_id (ref)",
  "file_name": "Chalisa_Sangrah.docx",
  "file_type": "docx | pdf | image",
  "file_url": "https://...",
  "status": "uploaded | parsing | parsed | published | error",
  "parsed_items_count": 12,
  "error_message": null,
  "parsed_data": {},
  "created_at": "2026-01-15T08:30:00Z",
  "published_at": null
}
```

app_settings

```
{
  "_id": "ObjectId",
  "key": "featured_content | home_banner | daily_shloka | announcement",
  "value": {},
  "is_active": true,
  "updated_at": "2026-01-15T08:30:00Z",
  "updated_by": "admin_user_id"
}
```


vedachat_conversations

```
{
  "_id": "ObjectId",
  "user_id": "user_profile_id (ref)",
  "messages": [
    {
      "role": "user",
      "content": "What does Krishna say about duty?",
      "timestamp": "2026-01-15T08:30:00Z"
    },
    {
      "role": "assistant",
      "content": "In Bhagavad Gita, Chapter 2, Verse 47...",
      "references": [
        {"book": "Bhagavad Gita", "chapter": 2, "verse": 47}
      ],
      "timestamp": "2026-01-15T08:30:05Z"
    }
  ],
  "created_at": "2026-01-15T08:30:00Z",
  "last_message_at": "2026-01-15T08:30:05Z"
}
```

Total Collections: 22

5. BACKEND API ENDPOINTS

5.1 Authentication APIs

Method	Endpoint	Description	Access
POST	/api/auth/admin/login	Admin login (email + password)	Public
POST	/api/auth/admin/register	Create new admin (Super Admin only)	Super Admin
POST	/api/auth/user/register	User registration (OTP/email)	Public
POST	/api/auth/user/login	User login	Public
GET	/api/auth/me	Get current user/admin profile	Authenticated
POST	/api/auth/refresh	Refresh JWT token	Authenticated

5.2 Content Management APIs (Admin)

Method	Endpoint	Description	Access
POST	/api/admin/upload/docx	Upload DOCX file for parsing	Content Admin+
POST	/api/admin/upload/pdf	Upload PDF for Vedas parsing	Content Admin+
POST	/api/admin/upload/publish/{upload_id}	Publish parsed content to DB	Content Admin+
GET	/api/admin/uploads	List all uploads with status	Content Admin+
GET	/api/admin/uploads/{id}	Get parsed preview of upload	Content Admin+
DELETE	/api/admin/uploads/{id}	Delete upload	Super Admin

5.3 Content Items APIs

Method	Endpoint	Description	Access
GET	/api/content/items	List all content items (filter by category, status)	Public
GET	/api/content/items/{id}	Get single content item with full details	Public
POST	/api/content/items	Create content item	Content Admin+
PUT	/api/content/items/{id}	Update content item	Content Admin+
DELETE	/api/content/items/{id}	Delete content item	Super Admin
PATCH	/api/content/items/{id}/status	Change status (draft/published/archived)	Content Admin+
PATCH	/api/content/items/{id}/sort	Update sort order	Content Admin+

5.4 Verse APIs

Method	Endpoint	Description	Access
GET	/api/content/items/{item_id}/verses	Get all verses for a content item	Public
GET	/api/content/verses/{id}	Get single verse with meanings	Public
PUT	/api/content/verses/{id}	Update verse (Sanskrit, transliteration)	Content Admin+
POST	/api/content/verses/{id}/meanings	Add/update meaning for a language	Content Admin+
GET	/api/content/verses/{id}/meanings/{lang}	Get meaning in specific language	Public

5.5 Katha & Puja APIs

Method	Endpoint	Description	Access
GET		List all kathas	Public

Method	Endpoint	Description	Access
	/api/katha/items		
GET	/api/katha/items/{id}	Get katha with intro, vidhi, samagri	Public
GET	/api/katha/items/{id}/chapters	Get katha chapters/verses	Public
POST	/api/katha/items	Create katha	Content Admin+
PUT	/api/katha/items/{id}	Update katha	Content Admin+

5.6 Arti APIs

Method	Endpoint	Description	Access
GET	/api/arti/items	List all artis	Public
GET	/api/arti/items/{id}	Get arti with lines	Public
POST	/api/arti/items	Create arti	Content Admin+
PUT	/api/arti/items/{id}	Update arti	Content Admin+

5.7 Divya Granth APIs

Method	Endpoint	Description	Access
GET	/api/granth/books	List all granth books	Public
GET	/api/granth/books/{id}	Get book with chapters	Public
GET	/api/granth/chapters/{id}	Get chapter with verses	Public
GET	/api/granth/chapters/{id}/verses	Get paginated verses	Public
POST	/api/granth/books	Create book	Super Admin
PUT	/api/granth/books/{id}	Update book	Content Admin+
PUT	/api/granth/verses/{id}	Update verse	Content Admin+

5.8 Vedas & Puranas APIs

Method	Endpoint	Description	Access
GET	/api/vedas/books	List all veda/purana books	Public
GET	/api/vedas/books/{id}	Get book with chapters	Public
GET	/api/vedas/chapters/{id}/verses	Get paginated verses	Public
POST	/api/vedas/books	Create book + trigger PDF parsing	Super Admin
PUT	/api/vedas/verses/{id}	Update verse	Content Admin+

5.9 VedaChat APIs

Method	Endpoint	Description	Access
POST	/api/vedachat/message	Send message, get AI response	Authenticated User
GET	/api/vedachat/conversations	Get user's conversation history	Authenticated User
GET	/api/vedachat/conversations/{id}	Get specific conversation	Authenticated User
DELETE	/api/vedachat/conversations/{id}	Delete conversation	Authenticated User

5.10 User APIs

Method	Endpoint	Description	Access
GET	/api/user/profile	Get user profile	Authenticated
PUT	/api/user/profile	Update profile (name, language, mode)	Authenticated
GET	/api/user/saved-shlokas	Get saved shlokas (daily routine)	Authenticated
POST	/api/user/saved-shlokas	Save shloka to routine	Authenticated
DELETE	/api/user/saved-shlokas/{id}	Remove from routine	Authenticated

Method	Endpoint	Description	Access
PUT	/api/user/saved-shlokas/reorder	Reorder saved shlokas	Authenticated
POST	/api/user/like/{item_id}	Like/unlike content	Authenticated
GET	/api/user/likes	Get all liked items	Authenticated
GET	/api/user/history	Get reading history	Authenticated
POST	/api/user/history	Update reading progress	Authenticated
GET	/api/user/videos	Get user-created videos	Authenticated

5.11 Home & Discovery APIs

Method	Endpoint	Description	Access
GET	/api/home/today	Get today's data (shloka, panchang, greeting)	Public
GET	/api/home/featured	Get featured content	Public
GET	/api/home/categories	Get category grid (8 tiles)	Public
GET	/api/search?q=...	Search across all content	Public

5.12 Admin Management APIs

Method	Endpoint	Description	Access
GET	/api/admin/users	List all app users	Content Admin+
GET	/api/admin/users/{id}	Get user details	Content Admin+
GET	/api/admin/admins	List all admin users	Super Admin
POST	/api/admin/admins	Create admin user	Super Admin
PUT	/api/admin/admins/{id}	Update admin role	Super Admin
DELETE	/api/admin/admins/{id}	Deactivate admin	Super Admin
GET	/api/admin/dashboard	Dashboard stats	Content Admin+
GET	/api/admin/panchang	List panchang entries	Content Admin+
POST	/api/admin/panchang	Create panchang entry	Content Admin+
PUT	/api/admin/panchang/{id}	Update panchang	Content Admin+

Method	Endpoint	Description	Access
GET	/api/admin/settings	Get app settings	Super Admin
PUT	/api/admin/settings/{key}	Update app setting	Super Admin

5.13 Audio & Media APIs

Method	Endpoint	Description	Access
POST	/api/media/generate-tts	Generate TTS audio for text	Content Admin+
POST	/api/media/generate-image	Generate shloka card image	Authenticated
POST	/api/media/generate-video	Generate shloka video	Authenticated
POST	/api/media/upload	Upload audio/image file	Content Admin+

Total API Endpoints: ~70+

6. ADMIN PANEL — SCREENS & FLOWS

6.1 Admin Panel Navigation (Sidebar)

```
SANATAN SAATHI ADMIN
├─ Dashboard (overview stats)
├─ Content Upload
│   ├─ Upload DOCX
│   ├─ Upload PDF (Vedas)
│   └─ Upload History
├─ Content Manager
│   ├─ Vedic Mantras
│   ├─ Chalisa
│   ├─ Ashtakam
│   ├─ Sahasranama
│   ├─ Nama Ramayanam
│   └─ All Content
├─ Verse Manager
│   └─ Edit verses, meanings, word breakdown
├─ Arti Manager
│   └─ Manage artis + lines
├─ Katha Manager
│   └─ Manage kathas + vidhi + samagri
├─ Granth Manager
│   ├─ Bhagavad Gita
│   ├─ Ramayana
│   └─ Mahabharata
├─ Vedas & Puranas
│   └─ Manage veda books + PDF parsing
├─ Audio Manager
│   ├─ Upload Audio
│   ├─ Generate TTS
│   └─ YouTube Links
├─ VedaChat Settings
│   └─ Context configuration
├─ Panchang Settings
│   └─ Daily data, festivals, panchak
├─ Users
│   └─ App user list + details
├─ Admin Users
│   └─ Manage admins (Super Admin only)
├─ App Settings
│   ├─ Featured Content
│   ├─ Home Banner
│   ├─ Daily Shloka
│   └─ Announcements
└─ My Profile
```


6.2 Key Admin Panel Screens

Dashboard

- Total content items by category (pie chart)
- Total users (daily/weekly/monthly growth)
- Content upload status (pending/published)
- Recent uploads
- Today's daily shloka preview
- Quick action buttons

Content Upload Screen

- Drag & drop DOCX file
- Select category (Chalisa, Vedic Mantra, etc.)
- Click "Parse with AI" button
- Preview parsed data in table format
- Edit any parsed row before publishing
- Error rows highlighted in red
- "Publish All" or "Publish Selected" buttons

Content Manager Screen

- Table view: Title, Category, Status, Verses Count, Sort Order, Actions
- Filter by: Category, Status, Language
- Search by title
- Inline status toggle (Draft/Published/Archived)
- Drag to reorder
- Click to edit full details

Verse Manager Screen

- Select content item from dropdown
- See all verses in table
- Click verse to expand editor
- Edit: Sanskrit text, Transliteration
- Tab for each language meaning
- Word breakdown editor (add/remove/edit words)
- Save individual verse or bulk save

Audio Manager Screen

- List all content items with audio status
- Upload audio file (drag & drop)
- Enter YouTube URL
- Generate TTS button (calls OpenAI TTS API)
- Audio preview player
- Set audio timestamps per verse

Panchang Settings Screen

- Calendar view showing existing entries
- Form: Date, Tithi, Nakshatra, Yoga, Karana, Sunrise, Sunset
- Festival name (Hindi + English)
- Link to specific content (for festival-related mantras)
- Panchak/Bhadra toggle
- Bulk import from CSV

6.3 Role-Based Access

Screen	Super Admin	Content Admin	Moderator
Dashboard	Full	Full	View only
Content Upload	Full	Upload + Parse + Publish	Upload + Parse (no publish)
Content Manager	Full CRUD + Delete	CRUD (no delete)	View + Flag
Verse Manager	Full	Full	View only
Audio Manager	Full	Full	View only
Katha Manager	Full	Full	View + Flag
Granth Manager	Full	Edit verses	View only
Vedas Manager	Full	Edit verses	View only
Panchang	Full	Full	View only
Users	Full + Subscription	View only	No access
Admin Users	Full CRUD	No access	No access
App Settings	Full	No access	No access
Payments	Full	No access	No access

7. AI INTEGRATIONS

All AI services use **Emergent LLM Key** (no additional API keys needed):

7.1 VedaChat (Claude API)

- **Model:** Claude Sonnet 4.5 (via Emergent Key)
- **Purpose:** Answer spiritual questions with scripture references
- **Context:** All published content from DB is used as context
- **Features:**
 - Multi-turn conversations
 - Scripture verse references (Book, Chapter, Verse)
 - Hindi + English responses based on user language preference
 - Conversation history stored in MongoDB

7.2 DOCX Parser (Claude API)

- **Model:** Claude Sonnet 4.5 (via Emergent Key)
- **Purpose:** Parse uploaded DOCX files into structured data
- **Extracts:**
 - H1 → Category
 - H2 → Item Name (title)
 - H3 → Section
 - Verse blocks → Individual verses with Sanskrit + transliteration
- **Output:** Structured JSON for preview and DB insertion

7.3 PDF Parser (Claude API)

- **Model:** Claude Sonnet 4.5 (via Emergent Key)
- **Purpose:** Extract shlokas from Vedas/Puranas PDFs
- **Process:** PDF → pdfplumber extracts text → Claude structures into verses

7.4 Text-to-Speech (OpenAI TTS)

- **Model:** OpenAI TTS (via Emergent Key)
- **Purpose:** Generate audio for verses in Hindi/Sanskrit

- **Voices:** Multiple voice options (male/female, Hindi/Sanskrit)
- **Output:** Audio file URL stored in DB

7.5 Image Generation (Gemini Nano Banana)

- **Purpose:** Generate shloka card backgrounds and decorative images
- **Use Case:** When user creates a shareable shloka card
- **Output:** Beautiful background image with Indian art style

7.6 Video Generation (Sora 2)

- **Purpose:** Generate short shloka video reels
 - **Use Case:** User creates video from a verse
 - **Output:** Short video with shloka text overlay
-

8. AUTHENTICATION & ROLE SYSTEM

8.1 Admin Authentication (JWT)

- Email + Password login
- Bcrypt password hashing
- JWT tokens (access + refresh)
- Role embedded in JWT payload
- Middleware validates role for each endpoint

8.2 User Authentication (JWT)

- Email/Phone + OTP login
- Google OAuth option (future)
- JWT tokens for mobile app
- Token refresh mechanism

8.3 Role Hierarchy

```
Super Admin (Level 1)
├─ Can manage everything
├─ Creates Content Admins & Moderators

Content Admin (Level 2)
├─ Can upload, edit, publish content
├─ Cannot manage admins or payments

Moderator (Level 3)
├─ Can upload and preview
├─ Cannot publish without review
├─ Can flag issues
```

9. CONTENT PIPELINE (DOCX/PDF PARSING)

9.1 DOCX Upload Flow

```
Step 1: Admin selects DOCX file
      ↓
Step 2: File uploaded to server
      ↓
Step 3: python-docx extracts raw text with heading levels
      ↓
Step 4: Claude API receives extracted text with prompt:
      "Parse this Hindu scripture document. Extract:
        - H1 as category
        - H2 as item title (in Hindi + English)
        - H3 as section headers
        - Verse blocks with Sanskrit text and transliteration
        Return as structured JSON"
      ↓
Step 5: Claude returns structured JSON
      ↓
Step 6: Preview table shown to admin
        (each row = one verse, with edit capability)
      ↓
Step 7: Admin reviews, edits if needed
      ↓
Step 8: Admin clicks "Publish"
      ↓
Step 9: Data saved to MongoDB collections:
        - content_items (one per H2)
        - content_verses (one per verse)
        - verse_meanings (one per verse per language)
      ↓
Step 10: Upload log updated with status "published"
```

9.2 PDF Upload Flow (Vedas/Puranas)

```
Step 1: Admin uploads PDF
      ↓
Step 2: pdfplumber extracts text page by page
      ↓
Step 3: Claude API structures text into:
        - Book metadata
        - Chapter divisions
```

```
- Individual verses with meanings
↓
Step 4: Preview shown to admin
↓
Step 5: Publish saves to veda_books, veda_chapters, veda_verses
```

10. MOBILE APP (EXPO) — INTEGRATION GUIDE

10.1 Expo Project Setup (For Your VS Code)

```
# Install Expo CLI
npm install -g expo-cli

# Create project
npx create-expo-app SanatanSaathi --template blank

# Install dependencies
npx expo install expo-router expo-av expo-image
npx expo install @react-navigation/native @react-navigation/bottom-tabs
npx expo install zustand @tanstack/react-query axios
npx expo install expo-notifications expo-clipboard expo-sharing
```

10.2 API Configuration

```
// config/api.js
const API_BASE_URL = "https://your-backend-url.com/api";

export const api = axios.create({
  baseURL: API_BASE_URL,
  headers: { "Content-Type": "application/json" }
});

// Add JWT token to all requests
api.interceptors.request.use((config) => {
  const token = useAuthStore.getState().token;
  if (token) config.headers.Authorization = `Bearer ${token}`;
  return config;
});
```

10.3 Screen-to-API Mapping

Expo Screen	API Endpoints Used
Home Tab	GET /api/home/today , GET /api/home/featured , GET /api/home/categories

Expo Screen	API Endpoints Used
Category List	GET /api/content/items?category=chalisa
Content Detail	GET /api/content/items/{id} , GET /api/content/items/{id}/verses
Verse with Meaning	GET /api/content/verses/{id}/meanings/{lang}
Katha List	GET /api/katha/items
Katha Detail	GET /api/katha/items/{id} , GET /api/katha/items/{id}/chapters
Arti List	GET /api/arti/items
Arti Player	GET /api/arti/items/{id}
Granth Library	GET /api/granth/books
Chapter Reader	GET /api/granth/chapters/{id}/verses
Vedas Home	GET /api/vedas/books
Vedas Reader	GET /api/vedas/chapters/{id}/verses
VedaChat	POST /api/vedachat/message
Profile	GET /api/user/profile
Saved Shlokas	GET /api/user/saved-shlokas
Like Content	POST /api/user/like/{item_id}
Generate TTS	POST /api/media/generate-tts
Create Video	POST /api/media/generate-video

10.4 Tab Navigation Structure

```

BottomTabs
├── Home (HomeStack)
│   ├── HomeScreen
│   ├── CategoryListScreen
│   ├── ContentDetailScreen
│   └── VerseReaderScreen
├── Vedic Mantras (VedicStack)
│   ├── VedicCategoriesScreen (8 tiles)
│   ├── ContentListScreen
│   └── ContentDetailScreen (Beginner/Expert)
├── Divya Granth (GranthStack)
│   └── GranthLibraryScreen

```

```
| | BookTOCScreen
| | ChapterReaderScreen
| — Veda & Puranas (VedasStack)
| | VedashomeScreen
| | BookDetailScreen
| | VerseReaderScreen
| — VedaChat (ChatStack)
|   ChatScreen
```

11. DEVELOPMENT PHASES & TIMELINE

What I Build Here (This Platform)

Phase	What	Sessions	Priority
Phase 1	Database schema (all 22 collections) + Seed data	1 session	P0
Phase 2	Admin Auth (JWT + 3 roles) + Admin CRUD	1 session	P0
Phase 3	Content APIs (items, verses, meanings)	1 session	P0
Phase 4	Admin Panel UI (Dashboard, Content Manager, Verse Manager)	1-2 sessions	P0
Phase 5	DOCX Upload + Claude parsing pipeline	1 session	P0
Phase 6	Katha, Arti, Granth, Vedas APIs	1 session	P1
Phase 7	VedaChat AI integration	1 session	P1
Phase 8	User APIs (profile, likes, saved, history)	1 session	P1
Phase 9	TTS, Image Gen, Video Gen APIs	1 session	P1
Phase 10	Admin Panel (Audio Manager, Panchang, Settings)	1 session	P2
Phase 11	Home & Discovery APIs	1 session	P2
Phase 12	Polish, testing, documentation	1 session	P2
	Total	~10-12 sessions	

What You Build in VS Code (Expo)

Phase	What	Estimated Time
E1	Expo setup + Tab navigation shell	1-2 days
E2	Home tab (greeting, shloka card, grid)	2-3 days
E3	Vedic Mantras screens (Beginner/Expert)	3-4 days

Phase	What	Estimated Time
E4	Audio player with verse highlighting	2-3 days
E5	Divya Granth reader	2-3 days
E6	Vedas & Puranas reader	2 days
E7	VedaChat screen	2-3 days
E8	Profile, Settings, Daily Routine	2-3 days
E9	Share, Video Create	2-3 days
E10	Polish + App Store submission	3-5 days
	Total	~25-35 days

12. SETUP INSTRUCTIONS FOR LOCAL DEVELOPMENT

12.1 Backend API (After downloading from this platform)

```
# Clone/download the code
cd backend

# Create virtual environment
python -m venv venv
source venv/bin/activate # Mac/Linux
venv\Scripts\activate    # Windows

# Install dependencies
pip install -r requirements.txt

# Set environment variables (.env file)
MONGO_URL=mongodb://localhost:27017
DB_NAME=sanatan_saathi
EMERGENT_LLM_KEY=your_key_here

# Run server
uvicorn server:app --host 0.0.0.0 --port 8001 --reload
```

12.2 Admin Panel Frontend

```
cd frontend

# Install dependencies
yarn install

# Set environment variables (.env)
REACT_APP_BACKEND_URL=http://localhost:8001

# Run
yarn start
```

12.3 Expo Mobile App (Your VS Code)

```
# Create Expo project
npx create-expo-app SanatanSaathi

# Configure API URL
# In your Expo app config, point to your deployed backend:
API_URL=https://your-deployed-backend.com/api

# Run on device
npx expo start
```

SUMMARY

Metric	Value
Database Collections	22
API Endpoints	70+
Admin Panel Screens	15+
AI Integrations	6 (Claude Chat, Claude Parser x2, OpenAI TTS, Nano Banana, Sora 2)
Admin Roles	3 (Super Admin, Content Admin, Moderator)
Content Categories	7 (Vedic Mantras, Chalisa, Ashtakam, Sahasranama, Katha, Arti, Nama Ramayanam)
Sacred Text Books	3+ (Gita, Ramayana, Mahabharata)
Veda Books	4+ (Rig, Sama, Yajur, Atharva)
Languages Supported	12+
Development Sessions (Web)	~10-12
Development Time (Mobile)	~25-35 days

This blueprint is designed for your review. Please confirm if this matches your vision, and let me know if any changes are needed before I start coding Phase 1.